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Requirements Engineering & UML Use-Case Modelling
Problem Statement #22 - Smart Cities, Transport & Logistics
Smart Parking Space Rental & License Validator

1. Problem Context & Overview

The proposed smart city parking system handles real-time parking-bay reservations, progressive hourly-rate calculation, vehicle-number-plate validation through gate cameras, and detection of unauthorized overstays. The primary stakeholders/actors are the Vehicle Owner and Parking Warden.

2. Functional Requirements

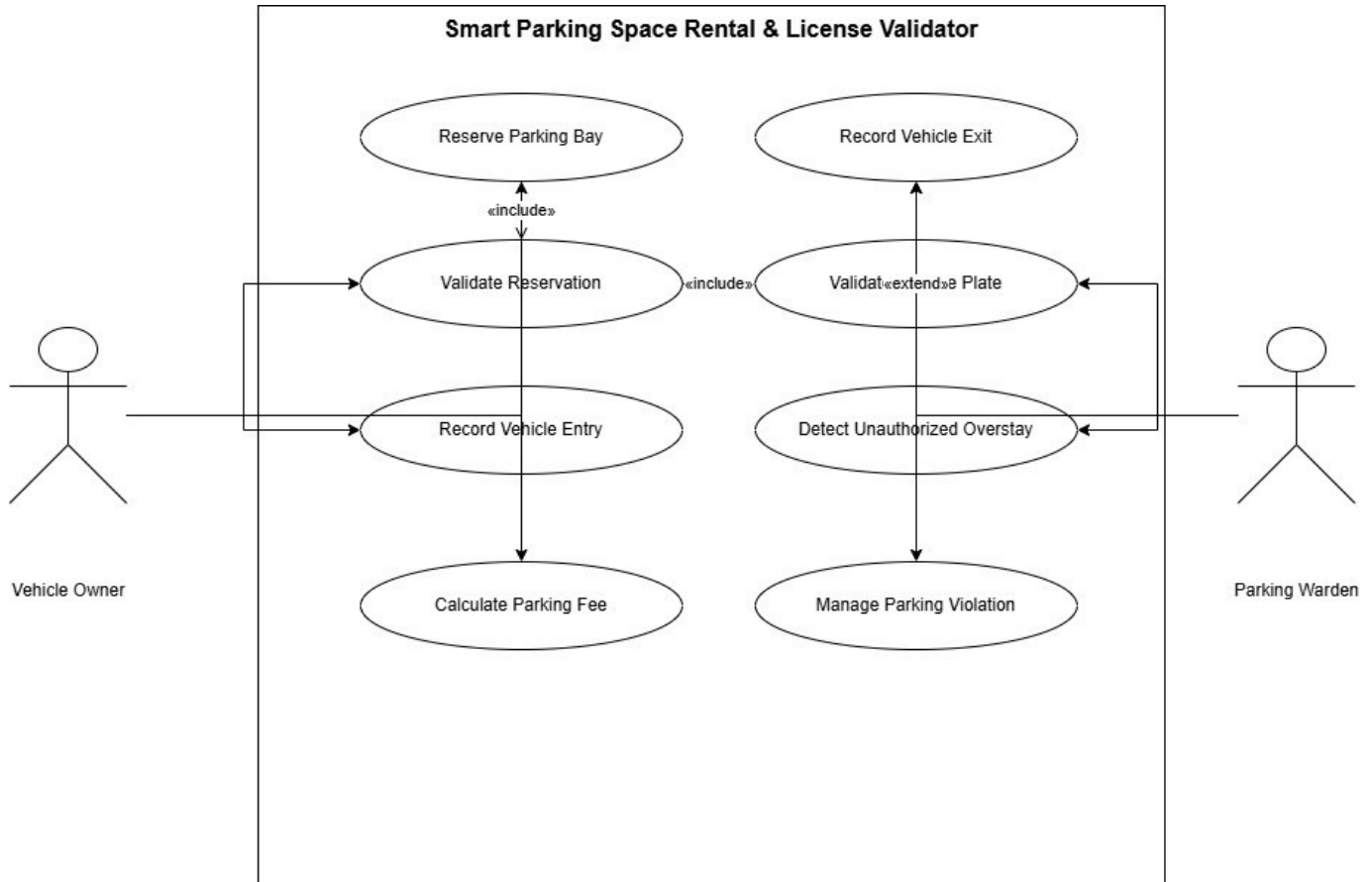
ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a Vehicle Owner to view available parking bays and reserve a bay for a selected parking period.	High	Pass: A valid available bay can be reserved and the reservation is stored. Fail: The system allows a reservation for an unavailable bay.	Enables real-time parking-space rental and prevents double booking.
FR-002	Functional	The system shall validate a vehicle's reservation by comparing the license plate captured by the gate camera with the active reservation.	High	Pass: A matching active reservation is accepted at the gate. Fail: A mismatched or expired reservation is rejected or	Ensures only authorized vehicles enter reserved parking.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
				flagged.	
FR-003	Functional	The system shall record vehicle entry and exit times using license-plate scans and calculate the parking fee using progressive hourly rates.	High	Pass: Entry/exit timestamps are recorded and the correct rate tier is applied. Fail: Incorrect duration or billing tier is produced.	Provides accurate usage tracking and billing.
FR-004	Functional	The system shall detect and flag vehicles that remain parked beyond their authorized reservation or parking period.	High	Pass: An overstay is identified after the permitted period and a violation is generated. Fail: An unauthorized overstay is not detected.	Supports enforcement of parking limits and reduces misuse of parking bays.
FR-005	Functional	The system shall allow a Parking Warden to view flagged parking violations and update the status of a violation after inspection.	Medium	Pass: The warden can view a flagged vehicle and record an appropriate violation status. Fail: A flagged violation cannot be reviewed or updated.	Provides operational control for parking enforcement.

3. Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The gate-camera license-plate lookup shall verify parking-reservation validity in under 300 ms and shall protect the validation request against unauthorized access.	High	Pass: Benchmarking under simulated peak load confirms a validation response below 300 ms while required security controls remain enabled. Fail: The target latency or security requirements are not met.	Fast and secure validation is required so gate processing does not create traffic delays or expose reservation data.
NFR-002	Reliability & Availability	The parking system shall maintain consistent reservation, entry/exit, billing, and violation records during normal operating conditions and recover without losing committed transactions after a temporary service interruption.	High	Pass: Recovery testing shows committed records remain intact and the service resumes correctly after simulated interruption. Fail: Committed parking or billing records are lost or corrupted.	Reliable records are essential for trustworthy reservations, billing, and enforcement.

4. UML Use-Case Diagram



5. Use-Case Flow Specification

Core Use Case: Reserve Parking Bay

Primary Actor: Vehicle Owner

Goal: Reserve an available parking bay for a selected period.

Preconditions: The Vehicle Owner can access the parking system; parking-bay information is available; and the requested parking period is valid.

Postconditions: A valid reservation is stored with the selected bay, vehicle/license-plate details, parking period, and reservation status. The bay is no longer shown as available for overlapping reservations.

Main Success Scenario:

1. Vehicle Owner opens the parking reservation function.
2. System displays currently available parking bays and applicable parking information.
3. Vehicle Owner selects a bay and specifies the desired parking period and vehicle number.
4. System checks that the selected bay is available for the requested period.
5. System creates and stores the reservation.
6. System confirms the reservation and associates it with the vehicle number.

Alternate Flow:

A1. At Step 4, the selected bay is no longer available for the requested period. The system rejects the reservation, informs the Vehicle Owner that the bay is unavailable, and presents available alternatives. The Vehicle Owner may select another bay and continue from Step 3.

6. Requirement Traceability Summary

FR-001 supports parking reservation. FR-002 supports gate-side license-plate reservation validation. FR-003 supports entry/exit recording and progressive billing. FR-004 supports unauthorized-overstay detection. FR-005 supports warden-side violation management. NFR-001 addresses the required gate validation performance and security target, while NFR-002 addresses reliable operation and record integrity.